Logarithms & Inverses of Exponential Functions

Review: Exponential Functions & Inverse Functions

- An exponential function is any function of the form $f(x) = P_0 a^x = P_0 e^{rt}$:
  - If $P_0 a^t = P_0 e^{rt}$, then $a = \mathbf{e^r}$ (Recall: $a^t = (e^r)^t = e^{rt}$).
  - P_0 is the **initial value** (the y-intercept $(0, P_0)$ of the graph).
  - If $a > 1$ (or $r > 0$), the function is strictly **increasing**.
  - If $0 < a < 1$ (or $r < 0$), the function is strictly **decreasing**.
- **Continuous Compounding:** A bank account with principal $\$A$ at continuous interest rate r yields balance Ae^{rt} after t years.
- **Invertibility Criterion:** A function is invertible if and only if it passes the **Horizontal Line Test (HLT)**. If $y = f(x)$ is invertible, then $f^{-1}(y) = x$.

Preliminaries: Inverting Exponentials

Question: Is every exponential function invertible? How do you know?

- **Board Answer:** *Yes! Because exponential functions are always strictly monotonic (always increasing or always decreasing everywhere), every horizontal line intersects the graph at most once. Therefore, they always pass the HLT!*

Recasting Doubling Time as an Inverse Function: To find the doubling time of an account paying 5% compounded continuously:

$$2A = Ae^{0.05t} \implies 2 = e^{0.05t}$$

If $P(t) = Ae^{0.05t}$, solving for t is equivalent to computing the inverse value: $\mathbf{t = P^{-1}(2A)}$.

Definition: The Natural Logarithm

If $f(x) = e^x$, we define its inverse function $f^{-1}(x)$ as the **natural logarithm**, denoted $\ln x$ (or $\log_e x$):

$$\ln \mathbf{x = c} \iff \mathbf{e^c = x}$$

Likewise, for base 10, the common logarithm is defined by $\log \mathbf{x = c} \iff \mathbf{10^c = x}$.

- **Domain and Range of $\ln x$:**

$$\text{Domain: } (\mathbf{0}, \infty) \quad \text{(strictly positive)}, \qquad \text{Range: } (-\infty, \infty) \quad \text{(all real numbers)}$$

- **Why does it make no sense to evaluate $\ln(0)$ or $\ln(x)$ for negative x?** Because $e^c > 0$ for all real numbers c. An exponential output is never zero and never negative!

Properties of Logarithms & Solving Exponential Equations

Core Algebraic Properties of Logarithms (Deducible from Exponent Laws):

$$\ln(AB) = \ln A + \ln B \qquad \log(AB) = \log A + \log B$$
$$\ln\left(\tfrac{A}{B}\right) = \ln A - \ln B \qquad \log\left(\tfrac{A}{B}\right) = \log A - \log B$$
$$\ln(A^p) = p\ln A \qquad \log(A^p) = p\log A$$
$$\ln(e^x) = x \quad \text{and} \quad e^{\ln x} = x \qquad \log(10^x) = x \quad \text{and} \quad 10^{\log x} = x$$

Applications: Growth Rates, Doubling Times, and Half-Lives

Problem 1: Exact Doubling Time at 5% Continuous Compounding

$$\begin{aligned} 2A &= Ae^{0.05t} \\ 2 &= e^{0.05t} \\ \ln(2) &= \ln\left(e^{0.05t}\right) \\ \ln 2 &= 0.05t \\ t &= \frac{\ln 2}{0.05} = \frac{0.693147\ldots}{0.05} \approx \mathbf{13.86} \text{ years} \end{aligned}$$

Problem 2: General Inversion of $P(x) = Ae^{rx}$

$$y = Ae^{rx} \implies \frac{y}{A} = e^{rx} \implies \ln\left(\frac{y}{A}\right) = rx \implies x = \frac{1}{r}\ln\left(\frac{y}{A}\right) = \frac{\mathbf{1}}{\mathbf{r}}(\ln \mathbf{y} - \ln \mathbf{A})$$

Thus: $\mathbf{P^{-1}(x)} = \frac{\mathbf{1}}{\mathbf{r}}(\ln \mathbf{x} - \ln \mathbf{A})$.

Solving Exponential Equations Analytically

Question: Solve $7^x = 2$ for x exactly:

$$\begin{aligned} 7^x &= 2 \\ \ln(7^x) &= \ln 2 \\ x \cdot \ln 7 &= \ln 2 \quad \text{using } \ln(A^p) = p\ln A \\ \mathbf{x} &= \frac{\ln \mathbf{2}}{\ln \mathbf{7}} \approx \frac{0.69315}{1.94591} \approx 0.3562 \end{aligned}$$

Key Takeaway for Exponential Modeling:
To solve for an exponent variable t algebraically, isolate the exponential base term and take the natural logarithm (ln) of both sides. The power property $\ln(a^t) = t\ln a$ pulls the variable down to linear form, allowing exact exact solution without guesswork.